LiteRise

Complete System Documentation & Implementation Guide

Adaptive English Learning Platform for Filipino Grade 3 Students

Version 2.0 - Complete Edition with Tutorial System
Technology Stack: React Native + MySQL + REST API
Development Timeline: 2 Weeks (14 Days)
Date: January 31, 2026

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Executive Summary

LiteRise is an adaptive English learning platform for Filipino Grade 3 students using Item Response Theory (IRT) for intelligent placement and a 3-phase learning flow (Lesson → Game → Quiz). The system includes Duolingo-style tutorials and walkthroughs to guide students through every step of their learning journey.

Core Features

- IRT-Based Placement (25 adaptive questions)
- Interactive onboarding tutorial (Duolingo-style)
- Step-by-step walkthroughs for first-time features
- 5 Modules × 13 Nodes = 65 learning units
- Adaptive branching: Quiz <70% triggers intervention
- Pre & Post Assessment with learning comparison
- Gamification: XP, badges, streaks with tutorials

Core Differentiators

- IRT Adaptive Testing: Unlike fixed assessments, every student receives questions perfectly matched to their ability level
- Intelligent Branching: System automatically detects struggling students and provides intervention with easier content
- Comprehensive Assessment: Pre-assessment placement followed by post-assessment to measure actual learning gains
- 3-Phase Learning: Lesson → Game → Quiz creates an engaging, effective learning experience
- MATATAG Aligned: All content follows Philippine Department of Education curriculum standards for Grade 3

System Overview & Philosophy

What is LiteRise?

LiteRise is an intelligent tutoring system that adapts to each student's ability level using Item Response Theory (IRT). The platform serves as a personalized English tutor that:

- Assesses student ability through adaptive placement testing
- Teaches concepts through multi-phase lessons (Lesson → Game → Quiz)
- Adapts content difficulty based on real-time performance
- Tracks comprehensive progress through 5 modules with 13 nodes each
- Motivates through gamification (XP, badges, streaks, leaderboards)
- Measures learning gains through pre and post assessments

Core Philosophy: Adaptive Learning

Every student learns at their own pace. This fundamental principle drives every decision in LiteRise:

- Struggling students receive easier content with more scaffolding (intervention)
- Advanced students receive challenging content with enrichment activities
- No student is left behind or held back from their potential
- Content difficulty adjusts automatically based on quiz performance
- Learning path is personalized to each student's ability level

Three-Phase Learning Flow

Each lesson follows a proven three-phase structure designed to maximize retention and engagement:

Phase	Activity	Purpose	Duration
1	LESSON Learn the Concept	Introduce new material with clear examples and explanations	5-8 min
2	GAME Practice Through Play	Reinforce learning through interactive, engaging gameplay	5-7 min
3	QUIZ Prove Mastery	Assess understanding and determine next steps	3-5 min
4	ADAPTIVE DECISION	Pass (≥70%): Continue to next lesson Fail (<70%): Intervention with easier content	—

Tutorial & Walkthrough System (Like Duolingo)

LiteRise includes a comprehensive tutorial system that guides students through every feature, ensuring they understand how to use the platform effectively. This prevents confusion and reduces drop-off rates.

Tutorial Design Principles

- Show, Don't Tell: Interactive demonstrations instead of long instructions
- Progressive Disclosure: Introduce features only when students need them
- Bite-Sized Steps: One concept at a time, max 3-4 screens per tutorial
- Hands-On Practice: Students try each feature immediately after learning it
- Skip Option: Advanced users can skip tutorials they don't need
- Replay Anytime: Tutorials accessible from Help menu for review

Tutorial Sequence

When	Tutorial	What Students Learn	Duration
First Launch	Welcome & Onboarding	App purpose, how adaptive learning works, what to expect	30 sec
Before Pre-Assessment	Placement Test Tutorial	Why we test, how it works (adaptive), no wrong answers, just do your best	45 sec
After Pre-Assessment	Understanding Your Level	What Beginner/Intermediate/Advanced means, personalized learning path	30 sec
First Dashboard Visit	Dashboard Tour	Module cards, progress bars, XP counter, streak, locked/unlocked nodes	60 sec
First Lesson (Node 1)	How Lessons Work	Read → Play Game → Take Quiz flow, time estimates, how to navigate	45 sec
First Game	Game Controls Tutorial	How to play, lives system, scoring, tap/drag mechanics	30 sec
First Quiz	Quiz Instructions	Read carefully, select answer, can't go back, passing score is 70%	30 sec
First Quiz Pass	Celebration & Progress	XP earned, next node unlocked, how progression works	20 sec
First Quiz Fail	Intervention Support	It's okay to struggle, we'll help with easier content, try again	30 sec
Complete Module 1	Module Completion	Congratulations, badge earned, next module unlocked	20 sec
After All Modules	Post-Assessment Intro	You've learned so much! Let's see your improvement, same test different questions	30 sec
After Post-	Comparison	How to read your improvement	45 sec

Assessment	Report Tour	chart, category gains, certificate	

Tutorial Implementation Details

Tutorials use modal overlays with spotlight effects (dim background, highlight target element) and animated mascot (optional: cute character like Duo the owl) to deliver instructions.

- Spotlight Effect: Dim entire screen except the element being explained
- Tooltips with Arrows: Point directly to buttons/features being described
- Animated Mascot (Optional): Friendly character that guides students (like Duo)
- Progress Dots: Show 'Step 2 of 4' so students know how long tutorial is
- Skip Button: Always visible in top-right corner
- Next/Got It Button: Large, colorful button to proceed
- Interactive Elements: Students must actually tap the button to continue (learning by doing)

Tutorial Database Schema

Track which tutorials students have seen to avoid repeating them:

Table: TutorialProgress

- StudentID - Foreign key to Students table
- TutorialName - e.g., 'welcome_onboarding', 'first_lesson', 'quiz_fail'
- CompletedAt - Timestamp when student completed tutorial
- Skipped - Boolean: Did student skip it?

MVP Requirements Checklist

Level 1: Foundation (Must Have)

Component	Requirements
Authentication	Registration, Login, JWT tokens, Session management
Database	Students, Modules, Nodes, AssessmentItems (500 questions), Sessions, Responses, Progress, Gamification, TutorialProgress tables
IRT Pre-Assessment	25-question adaptive test, Fisher Information selection, Theta calculation, Placement (Beginner/Intermediate/Advanced)
Learning Flow	View lesson → Take quiz → Score (Pass ≥70% or Fail <70%), First-time tutorials for each phase
Onboarding Tutorial	Welcome walkthrough, Dashboard tour, First lesson guide (Duolingo-style)

Level 2: Adaptive Intelligence (Core Feature)

Component	Requirements
Intelligent Branching	Quiz score analysis, Intervention for <70%, Normal progression for ≥70%, Intervention tutorial (supportive messaging)
Content Differentiation	3 difficulty levels per node: Beginner, Intermediate, Advanced
Module System	5 modules, 13 nodes each, Sequential unlocking, Module completion celebration
Post-Assessment	25 NEW questions, Same IRT algorithm, Learning comparison report, Post-assessment explanation tutorial

Level 3: Engagement (Important)

Dashboard with progress visualization (Dashboard tour tutorial), Basic gamification (XP, streaks, badges - with tooltip explanations), One game (Game controls tutorial), UX polish (loading states, error messages), Contextual help system with mascot/tooltips

Module Structure (65 Total Nodes)

LiteRise delivers 5 comprehensive modules aligned with the MATATAG Grade 3 English curriculum. Each module contains 13 nodes: 12 core lessons plus 1 final assessment, totaling 65 learning units.

Module 1: Grammar Fundamentals (BUILD FIRST ✓)

- Node 1: Nouns and Pronouns
- Node 2: Action Words (Verbs)
- Node 3: Describing Words (Adjectives)
- Node 4: Position Words (Prepositions)
- Node 5: Sentence Structure
- Node 6: Singular and Plural
- Node 7: Past and Present Tense
- Node 8: Question Formation
- Node 9: Sentence Types
- Node 10: Subject-Verb Agreement
- Node 11: Articles (a, an, the)
- Node 12: Conjunctions
- Node 13: Final Assessment

Module 2: Vocabulary Building (Post-MVP)

Common Words, Word Families, Synonyms/Antonyms, Homophones, Context Clues, Compound Words, Prefixes, Suffixes, Root Words, Idioms, Category Words, Multiple Meanings, Final Assessment

Module 3: Reading Comprehension (Post-MVP)

Main Idea, Details, Sequence, Cause/Effect, Compare/Contrast, Predictions, Conclusions, Character Analysis, Story Elements, Fact/Opinion, Author's Purpose, Inferences, Final Assessment

Module 4: Writing Skills (Post-MVP)

Complete Sentences, Capitalization, Punctuation, Paragraph Structure, Descriptive/Narrative/Opinion Writing, Writing Process, Editing, Dialogue, Letter Writing, Creative Expression, Final Assessment

Module 5: Listening & Speaking (Post-MVP)

Following Directions, Active Listening, Pronunciation, Phonics, Vowels, Blends/Digraphs, Rhyming, Syllables, Oral Presentation, Conversation, Storytelling, Listening Comprehension, Final Assessment

IRT System (Item Response Theory)

Item Response Theory (IRT) is the mathematical foundation that makes LiteRise adaptive. Unlike traditional fixed tests, IRT selects questions based on the student's current ability level, providing maximum accuracy with minimum questions.

The IRT Formula (3-Parameter Logistic Model)

$P(\theta) = c + (1 - c) / (1 + e^{(-a(\theta - b))})$

Parameter	Meaning	Range
θ (theta)	Student ability	-3.0 to +3.0
a	Discrimination (how well question separates abilities)	0.5 to 2.5
b	Difficulty (higher = harder)	-3.0 to +3.0
c	Guessing (probability of random correct)	0.25

Theta Update Algorithm

After each answer, the student's ability (theta) is updated using gradient ascent:

$\Delta\theta = \alpha \times a \times (y - P(\theta))$

Where $\alpha=0.3$ (learning rate), $y=1$ if correct/0 if incorrect

Placement Levels

- $\theta < -0.5 \rightarrow$ Beginner (struggles with grade-level)
- $-0.5 \leq \theta < 0.5 \rightarrow$ Intermediate (average performance)
- $\theta \geq 0.5 \rightarrow$ Advanced (excels at grade-level)

Fisher Information - Question Selection

To select the best next question, the system calculates Fisher Information, which measures how much we can learn about the student's ability from each question:

$I(\theta) = a^2 \times P(\theta) \times (1 - P(\theta))$

The question with the highest Fisher Information is selected. This typically occurs when $P(\theta) \approx 0.5$, meaning the question is perfectly matched to the student's ability - not too easy, not too hard.

Assessment System

Pre-Assessment (IRT Placement Test)

The initial placement test uses Item Response Theory (IRT) to determine each student's ability level with maximum accuracy in just 25 questions.

How It Works:

- Step 1 - Initialization: Student begins with neutral ability ($\theta = 0.0$)
- Step 2 - Question Selection: System calculates Fisher Information for all unused questions and selects the one with highest information value
- Step 3 - Student Response: Student answers the selected question
- Step 4 - Theta Update: System updates ability estimate using gradient ascent
- Step 5 - Repeat: Steps 2-4 repeat for 25 questions (5 per category)
- Step 6 - Placement: Final theta determines placement level

Post-Assessment & Learning Comparison

After students complete all 5 modules (65 nodes total), they take a post-assessment to measure their learning gains. This comprehensive evaluation compares their current ability against their initial placement to demonstrate real, measurable progress.

Post-Assessment Structure:

- 25 Questions: Same adaptive IRT methodology as pre-assessment, but with different questions
- Same Categories: 5 questions each from Grammar, Vocabulary, Reading, Writing, and Listening/Speaking
- Adaptive Selection: Questions still selected based on Fisher Information
- Fresh Question Pool: All questions are different from pre-assessment to prevent memorization

Technology Architecture

Technology Stack

Component	Technology	Purpose
Frontend	React Native	Cross-platform mobile app (iOS & Android)
Backend API	Node.js + Express	RESTful API server
Database	MySQL	Relational data storage
Authentication	JWT	Secure session management

Success Criteria & Testing

MVP Launch Criteria

- ✓ Welcome onboarding tutorial shows on first launch
- ✓ Pre-assessment tutorial explains adaptive testing
- ✓ Dashboard tour highlights all key features
- ✓ First lesson/game/quiz have step-by-step guides
- ✓ Intervention tutorial provides supportive messaging
- ✓ Help button accessible throughout app
- ✓ All tutorials can be skipped but are tracked
- ✓ Tooltips explain XP, streaks, badges
- ✓ Post-assessment tutorial shows improvement comparison
- ✓ Students complete onboarding in <5 minutes

Key Metrics to Monitor

- Placement test completion time: 10-15 minutes
- Quiz pass rate: 60-80% (indicates appropriate difficulty)
- Intervention rate: 20-30% (shows adaptive system is working)
- Average theta improvement: $\Delta\theta \geq 0.5$ (demonstrates learning)
- Student retention: ≥70% complete at least 10 nodes

Implementation Checklist

Database Setup

- Create TutorialProgress table
- Seed with tutorial names (welcome, dashboard_tour, first_lesson, etc.)
- Create all 11 tables with proper relationships
- Seed 250 pre-assessment questions with calibrated a, b, c parameters
- Seed 250 post-assessment questions (different from pre-test)
- Create Module 1 with 13 nodes × 3 difficulty levels
- Add indexes on StudentID, NodeID, SessionID for performance

Backend Development

- GET /api/tutorial/check endpoint
- POST /api/tutorial/complete endpoint
- Implement JWT authentication (register, login, verify)
- Build IRT engine (probability calc, Fisher Info, theta update)
- Create placement test endpoints (start, next-question, submit, complete)
- Build learning endpoints (modules, nodes, lessons, quiz, submit)
- Implement adaptive branching logic (intervention trigger)
- Create post-assessment and comparison endpoints

Frontend Development

- Tutorial modal component with spotlight effect
- Animated mascot character (optional)
- Progress dots (Step X of Y)
- Skip and Next buttons
- Welcome onboarding (4 screens)
- Dashboard tour (3-4 highlights)
- Lesson/Game/Quiz tutorials
- Intervention support message
- Floating help button
- Tooltip system for XP/streaks/badges
- Post-assessment explanation
- Comparison report tour
- Registration and login screens
- Pre-assessment test interface (25 questions)
- Dashboard with module cards and progress visualization
- Lesson viewer (displays content by student level)
- Quiz interface (10 questions per node)
- Intervention flow (easier content → retry quiz)
- Post-assessment interface

- Comparison report screen with charts

Testing Scenarios

- Complete registration → Pre-assessment → Placement assignment
- Pass quiz (≥70%) → Next node unlocks
- Fail quiz (<70%) → Intervention triggered → Retry successful
- Complete all 65 nodes → Post-assessment appears
- Post-assessment → Comparison shows improvement
- Test with actual Grade 3 students for age-appropriateness
- All tutorials can be skipped and tracked
- Tutorial flow works smoothly end-to-end

Conclusion

With this comprehensive documentation, you have everything needed to build LiteRise MVP in 2 weeks. The tutorial system, inspired by Duolingo, ensures that Grade 3 students can use the platform effectively from day one. Focus on Level 1 and Level 2 features, ensure the IRT system works correctly, and test thoroughly with real students.

What Makes LiteRise Special:

- IRT Adaptive Testing: Every student gets questions perfectly matched to their ability level
- Intelligent Branching: System automatically detects struggles and provides intervention
- Comprehensive Assessment: Pre and post testing measures real learning gains
- 3-Phase Learning: Lesson → Game → Quiz creates engaging, effective learning
- Tutorial System: Duolingo-style walkthroughs ensure students understand every feature
- MATATAG Aligned: Content follows Philippine DepEd curriculum for Grade 3

Critical Success Factors:

- Read relevant skills documentation before implementing each feature
- Understand IRT mathematics thoroughly before implementation
- Test the IRT engine independently before integration
- Prioritize database design and proper question calibration
- Implement all tutorials following the Duolingo-style principles
- Conduct user testing with actual Grade 3 students
- Monitor performance metrics from Day 1
- Ensure tutorial flow is smooth and non-intrusive

Good luck with implementation! □